# Solution for Chapter 5

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## 5.1

(a) When $p = 2$,

$$\mathcal{L}(\mathbf{W}, b, \alpha, \beta, \xi) = \frac{1}{2}||\mathbf{w}||^2 + C\sum_{i=1}^{m}\xi_i^2 - \sum_{i=1}^{m}\alpha_i[y_i(\mathbf{w}\mathbf{x}_i + b) - 1 + \xi_i] - \sum_{i=1}^{m}\beta_i\xi_i$$

Setting the gradient of the Lagrangian with respect to the $\mathbf{w}, b, \xi_i$ is to zero, the results we got is same as (5.26), (5.27), (5.29), (5.30). (5.28) is replaced by $2C\xi_i = \alpha_i + \beta_i$.

$$\begin{aligned}\inf \mathcal{L} &= \frac{1}{2}||\sum_{i=1}^{m}\alpha_i y_i \mathbf{x}_i||^2 + C\sum_{i=1}^{m}(\frac{\alpha_i + \beta_i}{2C})^2 - \sum_{i=1}^{m}\alpha_i y_i b + \sum_{i=1}^{m}\alpha_i - \sum_{i=1}^{m}\alpha_i\xi_i - \sum_{i=1}^{m}\beta_i\xi_i - \sum_{i=1}^{m}\alpha_i y_i \mathbf{w}\mathbf{x} \\ &= \sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\alpha_i\alpha_j y_i y_j \mathbf{x}_i\mathbf{x}_j - \frac{1}{4}\sum_{i=1}^{m}\frac{(\alpha_i + \beta_i)^2}{C}\end{aligned}$$

(b) Yes, this problem is still convex because the first two items are convex, and $-\frac{1}{4}\sum_{i=1}^{m}\frac{(\alpha_i+\beta_i)^2}{C}$ is concave but we can use convec techniques to solve it.

## 5.2

## 5.3

Add weight to Lagrangian, get

$$\mathcal{L}(\mathbf{W}, b, \alpha, \beta, \xi) = \frac{1}{2}||\mathbf{w}||^2 + C\sum_{i=1}^{m}\xi_i p_i - \sum_{i=1}^{m}\alpha_i[y_i(\mathbf{w}\mathbf{x}_i + b) - 1 + \xi_i] - \sum_{i=1}^{m}\beta_i\xi_i$$

For gradient, (5.26)(5.27)(5.29)(5.30) is same as regular case. (5.28) is replaced with $Cp_i - \alpha_i - \beta_i$. Put them into Lagrangian, get the same result with regular case.

$$\inf \mathcal{L} = \sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\alpha_i\alpha_j y_i y_j \mathbf{x}_i\mathbf{x}_j$$

The constraint condition is $\sum_{i=1}^{m}\alpha_i y_i = 0 \wedge 0 \leq \alpha_i, \beta_i \leq Cp_i$

## 5.4

(a)

$$
\begin{aligned}
Equation(5.33) &= \alpha_1 + \alpha_2 + \sum_{i=3}^{m} \alpha_i \\
&- \frac{1}{2}[\alpha_1\alpha_1 y_1 y_1 (\mathbf{x}_1)\mathbf{x}_1 + \alpha_1\alpha_2 y_1 y_2 (\mathbf{x}_1\mathbf{x}_2) + \sum_{i=3}^{m} \alpha_1\alpha_i y_1 y_i (\mathbf{x}_1\mathbf{x}_i) \\
&+ \alpha_2\alpha_1 y_2 y_1 (\mathbf{x}_2\mathbf{x}_1) + \alpha_2\alpha_2 y_2 y_2 (\mathbf{x}_2\mathbf{x}_2) + \sum_{i=3}^{m} \alpha_2\alpha_i y_2 y_i (\mathbf{x}_2\mathbf{x}_i)] \\
&= \alpha_1 + \alpha_2 - \frac{1}{2}K_{11}\alpha_1^2 - \frac{1}{2}K_{22}\alpha_2^2 - sK_{12}\alpha_1\alpha_2 - y_1\alpha_1 v_1 - y_2\alpha_2 v_2
\end{aligned}
$$

For constraint condition,

$$
\begin{aligned}
0 = \sum_{i=1}^{m} \alpha_i y_i &= \alpha_1 y_1 + \alpha_2 y_2 + \sum_{i=3}^{m} \alpha_i y_i \\
&= \alpha_1 y_1^2 + \alpha_2 y_2 y_1 + y_1 \sum_{i=3}^{m} \alpha_i y_i \\
&= \alpha_1 + s\alpha_2 + \gamma = 0
\end{aligned}
$$

**5.6**

(a) First item in original optimization problem is $||\mathbf{w}||^2$,

$$
\begin{aligned}
||\mathbf{w}||^2 &= ||\sum_{i=1}^{m} \alpha_i y_i \mathbf{x}_i||^2 \\
&= (\alpha_1 y_1 \mathbf{x}_1 + \alpha_2 y_2 \mathbf{x}_2 + \cdots + \alpha_m y_m \mathbf{x}_m)(\alpha_1 y_1 \mathbf{x}_1 + \alpha_2 y_2 \mathbf{x}_2 + \cdots + \alpha_m y_m \mathbf{x}_m)
\end{aligned}
$$

To make above equation equal to $\sum_{i=1}^{m} \alpha_i^2$, following should holds

$$
\mathbf{x}_i\mathbf{x}_j = 0, i \neq j
$$

.

(b)

$$L = \frac{1}{2}\sum_{i=1}^{m}\alpha_i^2 + C\sum_{i=1}^{m}\xi_i - \sum_{i=1}^{m}\beta_i \cdot [y_i(\sum_{j=1}^{m}\alpha_j y_j \mathbf{x}_i \cdot \mathbf{x}_j + b) - 1 + \xi_i] - \sum_{i=1}^{n}\gamma_i\xi_i - \sum_{i=1}^{m}\varepsilon_i\alpha_i$$

KKT condition:

$$\nabla_{\alpha_i} L = \alpha_i - \sum_{j=1}^{m}\beta_j y_j y_i \mathbf{x}_i \mathbf{x}_j - \varepsilon_i = 0$$

$$\nabla_b L = \sum_{i=1}^{m}\beta_i \cdot y_i = 0$$

$$\nabla_{\xi_i} L = C - \beta_i - \gamma_i = 0$$

$$\beta_i[y_i(\sum_{j=1}^{m}\alpha_j y_j \mathbf{x}_i \mathbf{x}_j + b) - 1 + \xi_i] = 0$$

$$\gamma_i \cdot \xi_i = 0$$

$$\varepsilon_i \cdot \alpha_i = 0$$

(c)

$$L = \sum_{i=1}^{m}||\alpha_i|| + C\sum_{i=1}^{m}\xi_i - \sum_{i=1}^{m}\beta_i \cdot [y_i(\sum_{j=1}^{m}\alpha_j y_j \mathbf{x}_i \cdot \mathbf{x}_j + b) - 1 + \xi_i] - \sum_{i=1}^{n}\gamma_i\xi_i - \sum_{i=1}^{m}\varepsilon_i\alpha_i$$

KKT condition:

$$\nabla_{\alpha_i} L = 1 - \sum_{j=1}^{m}\beta_j y_j y_i \mathbf{x}_i \mathbf{x}_j - \varepsilon_i = 0$$

$$\nabla_b L = \sum_{i=1}^{m}\beta_i \cdot y_i = 0$$

$$\nabla_{\xi_i} L = C - \beta_i - \gamma_i = 0$$

$$\beta_i[y_i(\sum_{j=1}^{m}\alpha_j y_j \mathbf{x}_i \mathbf{x}_j + b) - 1 + \xi_i] = 0$$

$$\gamma_i \cdot \xi_i = 0$$

$$\varepsilon_i \cdot \alpha_i = 0$$

**5.7**

(a) Because $\mathbf{x}_1, \ldots, \mathbf{x}_d$ can be shattered, there always exists $\mathbf{w}$ which can holds $\mathbf{w}\mathbf{x}_x = y_i$.

$$\begin{aligned} 1 &\leq |\mathbf{w}\mathbf{x}_i| \\ &= |\mathbf{w}y_i\mathbf{x}_i| \\ &= \mathbf{w}y_i\mathbf{x}_i \end{aligned}$$

Add above for $i = 1, ..., d$, get

$$
\begin{aligned}
d &\leq \sum_{i=1}^{d} \mathbf{w} y_i \mathbf{x}_i \\
&\leq \mathbf{w} \sum_{i=1}^{d} y_i \mathbf{x}_i \\
d &\leq ||\mathbf{w}|| || \sum_{i=1}^{m} y_i \mathbf{x}_i || \\
&\leq \Lambda || \sum_{i=1}^{m} y_i \mathbf{x}_i ||
\end{aligned}
$$

(b) According to Jesen's inequality, $E(x)^2 \leq E(x^2)$, so $E(x) \leq \sqrt{E(x^2)}$.

$$
\begin{aligned}
d &\leq \Lambda E[|| \sum_{i=1}^{m} y_i \mathbf{x}_i ||] \\
&\leq \Lambda \sqrt{E[|| \sum_{i=1}^{m} y_i \mathbf{x}_i ||^2]} \\
&= \Lambda \sqrt{\sum_{i=1}^{d} ||\mathbf{x}_i||^2}
\end{aligned}
$$

(c)

$$
\begin{aligned}
d &\leq \Lambda \sqrt{\sum_{i=1}^{d} ||\mathbf{x}_i||^2} \\
&\leq \Lambda \sqrt{d r^2}
\end{aligned}
$$

Solve the inequality, get $d \leq r^2 \Lambda^2$.